Assignment 1: Sources of Performance Inefficiency

Fall 2026

1 Scenario

During this assignment, you'll play the role of a performance engineer at a market maker. Your job is to analyze and critique the application included with this assignment. This application takes a stream of market data and implements your firm's trading strategy, sending orders to the exchange based on the messages received from it.

Since you're a member of the engineering division, you're primarily concerned with the performance of the application. In particular, you are not concerned with improving the strategy your firm has chosen to implement. For the purposes of this assignment, you can assume that decisions about the trading strategy have already been finalized by another division. Your goal is simply to minimize the tick-to-trade latency of that strategy's implementation. You shouldn't change, for instance, the fair value, the credit your firm requires, or the risk handling procedures. The application is shipped with a verification script that ensures the business logic remains unchanged for the set of scenarios we're using to test your application. This verification can be run using `make verify`.

The codebase is split into two main parts.

The `sim` directory holds the portion of the simulation that represents the exchange and the network that connects your application to it. You should not have to change anything in this directory for the purposes of this assignment, but you are permitted to reference it (for example, to view API details).

The `app` directory contains the application you're evaluating. This includes the `market_maker` implementing the strategy and the `evaluation` measuring its performance. The entry point for the simulation is `main.cpp`. The simulation interacts with the `market_maker` by calling handlers in `market_maker.cpp`. Your task is to submit a 1-page critique (details below), then discuss the application and your critique in a 15-minute interview. While you are not required to submit an updated application, you are allowed to make changes (to both the `market_maker` and `evaluation`) to provide any metrics you feel might support your assertions.

2 Critique

Included with the application is a design document that details the design decisions that were made and the rationale behind them. Some of those design decisions are good choices that

improve performance when it matters, while others are bad choices that hurt performance in unintended ways. To simplify this assignment, we ask that you limit your critique to design decisions specifically discussed in that document. Your critique should contain the following:

- Summarize one possible execution of the hotpath.
- A discussion of a bad design decision made in the `evaluation`, and a proposed correction.
- A discussion of a good design decision made in the `market_maker`.
- A discussion of two bad design decisions made in the `market_maker`, and proposed corrections.

Please use the provided `critique.tex` format and submit the resulting PDF. You are permitted (and encouraged) to collaborate with other students and use AI tools to help you understand the application, evaluate proposed changes, etc. That said, you should write the final 1-page critique yourself, and the interview will be conducted without any outside help, so take time to make sure that you also understand the application and the performance implications of your proposals.

Begin your submission with a short statement describing how you collaborated with other students, which AI tools you used, and how you used them. This statement does not count against the 1-page limit for the critique.